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CO4-ASSESSMENT TOOL - 01

Title: AI-Powered Personalized Nutrition and Health Recommendation Platform

1.Objective

The objective of testing is to verify that the AI-powered nutrition platform correctly collects user health information, analyzes nutritional requirements, generates personalized meal plans, monitors dietary intake, integrates wearable devices, provides health recommendations, and tracks user progress.
Testing also ensures that the system is **reliable, secure, usable, accurate, and responsive** under normal, boundary, invalid, and exceptional conditions.

2.Functional Requirements to be Tested

Req. ID	Functional Requirement
FR01	User registration and login
FR02	Collection and validation of personal health information
FR03	Collection of dietary preferences and restrictions
FR04	Collection of fitness goals
FR05	AI-based analysis of health and nutritional data
FR06	Generation of personalized meal plans
FR07	Monitoring and recording of dietary intake
FR08	Wearable device integration
FR09	Health and nutrition recommendations
FR10	Nutrition progress tracking
FR11	Updating user health information
FR12	Notifications and reminders
FR13	Secure handling of user information
FR14	Error handling and recovery

3. Non-Functional Requirements to be Tested

NFR ID	Requirement
NFR01	Performance
NFR02	Usability
NFR03	Reliability
NFR04	Security

NFR05	Compatibility
NFR06	Scalability
NFR07	Availability
NFR08	Accuracy

4. Test Scenarios

Scenario ID	**Test Scenario**
TS01	User registration
TS02	User login
TS03	Enter personal health information
TS04	Validate age, height and weight
TS05	Enter medical conditions
TS06	Select dietary preferences
TS07	Select fitness goal
TS08	Generate personalized nutrition plan
TS09	Record daily food intake
TS10	Calculate/display nutritional intake
TS11	Connect wearable device
TS12	Synchronize wearable data
TS13	Generate health recommendations
TS14	Track nutrition progress
TS15	Update health information
TS16	Receive reminders/notifications
TS17	Handle invalid/missing information
TS18	Handle unavailable AI/wearable services
TS19	Verify data security
TS20	Verify system performance and usability

5. Detailed Test Cases

Assumption: For test data, reasonable validation ranges are used for demonstration. Actual limits should follow the final system specification.

TC ID	**Test Scenario**	**Test Steps**	**Test Data**	**Expected Result**	**Actual Result**	**Status**
TC01	User registration	Open registration → enter details → submit	Name: Ravi, Email: ravi@gmail.com, Password: Ravi@123	Account is created successfully	To be executed	Not Executed
TC02	Registration with invalid email	Enter registration details with invalid email → submit	ravi@	Validation message is displayed	To be executed	Not Executed
TC03	Registration with existing email	Enter already registered email → submit	Existing email	System displays “Email already registered”	To be executed	Not Executed

TC04	Login with valid credentials	Enter email and password → click Login	Valid email/password	User successfully logs in	To be executed	Not Executed
TC05	Login with invalid password	Enter valid email and wrong password	Password: wrong123	Login is rejected with error message	To be executed	Not Executed
TC06	Login with blank fields	Leave email/password blank → Login	Blank values	Required-field messages are displayed	To be executed	Not Executed
TC07	Enter valid health information	Open health profile → enter details → save	Age: 25, Height: 170 cm, Weight: 65 kg	Information is saved successfully	To be executed	Not Executed
TC08	Age boundary – minimum	Enter minimum allowed age	Age: 18	Age is accepted	To be executed	Not Executed
TC09	Age below minimum	Enter age below allowed limit	Age: 17	System rejects value	To be executed	Not Executed
TC10	Invalid age	Enter non-numeric age	Age: abc	Validation error is displayed	To be executed	Not Executed
TC11	Height boundary	Enter minimum/maximum allowed height	Height: 100 cm / 250 cm	Valid boundary values are accepted	To be executed	Not Executed
TC12	Invalid height	Enter negative height	Height: -170 cm	System rejects value	To be executed	Not Executed
TC13	Weight boundary	Enter valid minimum/maximum weight	Weight: 30 kg / 200 kg	Valid weight is accepted	To be executed	Not Executed
TC14	Invalid weight	Enter negative weight	Weight: -60 kg	Validation error is displayed	To be executed	Not Executed
TC15	Missing health information	Leave mandatory health fields empty → Save	Age/weight blank	System asks user to enter required information	To be executed	Not Executed
TC16	Enter medical condition	Select medical condition → save	Diabetes	Condition is stored in health profile	To be executed	Not Executed
TC17	Dietary preference	Select vegetarian diet	Vegetarian	Meal recommendations contain vegetarian foods	To be executed	Not Executed
TC18	Food allergy	Select food allergy	Peanut allergy	Recommended meals exclude peanuts	To be executed	Not Executed
TC19	Fitness goal	Select weight-loss goal	Weight loss	Nutrition plan is generated according to goal	To be executed	Not Executed
TC20	Generate nutrition plan	Complete health profile → click Generate Plan	Valid health/profile data	Personalized meal plan is generated	To be executed	Not Executed
TC21	Generate plan with incomplete data	Leave mandatory profile information blank → Generate	Missing weight	System requests missing information	To be executed	Not Executed
TC22	Meal plan personalization	Change dietary preference → regenerate plan	Vegan	New plan contains suitable vegan meals	To be executed	Not Executed
TC23	Record food intake	Select food → enter quantity → save	Rice: 200 g	Food intake is recorded	To be executed	Not Executed
TC24	Invalid food quantity	Enter negative quantity	-200 g	System rejects quantity	To be executed	Not Executed
TC25	Daily nutrition calculation	Add multiple food items	Breakfast + lunch + dinner	Total calories/nutrients are calculated correctly	To be executed	Not Executed

TC26	Edit food entry	Open recorded meal → edit quantity → save	200 g → 250 g	Updated quantity and nutrition values are displayed	To be executed	Not Executed
TC27	Delete food entry	Select food entry → Delete	Lunch item	Entry is removed and totals are recalculated	To be executed	Not Executed
TC28	Connect wearable	Open wearable integration → connect device	Supported smartwatch	Device is connected successfully	To be executed	Not Executed
TC29	Unsupported wearable	Attempt to connect unsupported device	Unsupported device	System displays compatibility message	To be executed	Not Executed
TC30	Wearable synchronization	Connect device → Sync	Steps: 8,000	Latest wearable data is synchronized	To be executed	Not Executed
TC31	Wearable unavailable	Disconnect internet → Sync	No network	System displays synchronization failure and does not corrupt data	To be executed	Not Executed
TC32	Health recommendation	Complete health profile → request recommendation	Valid profile	Relevant health/nutrition recommendations are displayed	To be executed	Not Executed
TC33	Recommendation after profile update	Change weight/goal → request recommendation	New weight/goal	Recommendation reflects updated information	To be executed	Not Executed
TC34	Track progress	Record meals for multiple days → open progress	7 days of nutrition data	Progress chart/report is displayed correctly	To be executed	Not Executed
TC35	No progress data	Open progress without recorded data	No records	System displays appropriate “No data” message	To be executed	Not Executed
TC36	Update profile	Change health information → save	Weight: 65 → 68 kg	Updated information is stored	To be executed	Not Executed
TC37	Notification	Set meal reminder → wait for scheduled time	Reminder: 1:00 PM	Notification is generated at scheduled time	To be executed	Not Executed
TC38	Invalid reminder time	Enter invalid reminder time	25:90	System rejects invalid time	To be executed	Not Executed
TC39	Session security	Login → logout → use browser Back	Valid account	Protected pages cannot be accessed after logout	To be executed	Not Executed
TC40	Unauthorized access	Try opening another user's profile	Unauthorized account	Access is denied	To be executed	Not Executed
TC41	AI service failure	Request meal plan while AI service unavailable	Valid profile	Friendly error is displayed; system remains stable	To be executed	Not Executed
TC42	Database/service failure	Save health data during service interruption	Valid data	Error is handled without data corruption	To be executed	Not Executed
TC43	Performance	Generate recommendation under normal load	Valid profile	Recommendation is returned within specified response time	To be executed	Not Executed
TC44	Multiple users	Simulate many users accessing system	Multiple concurrent users	System remains responsive and stable	To be executed	Not Executed

6. Equivalence Partitioning

Equivalence Partitioning divides input data into valid and invalid groups.

Example: Age

Partition	**Test Data**	**Expected Result**
Valid	18–100	Accepted
Invalid – below range	0–17	Rejected
Invalid – negative	-1, -20	Rejected
Invalid – non-numeric	abc	Rejected
Empty	Blank	Required-field error

Example: Weight

Partition	**Test Data**	**Expected Result**
Valid	30–200 kg	Accepted
Below range	<30 kg	Rejected
Above range	>200 kg	Rejected
Negative	-50 kg	Rejected
Non-numeric	abc	Rejected
Empty	Blank	Required-field error

7. Boundary Value Analysis

Boundary values are especially important for health-profile inputs.

Input	**Boundary Value**	**Test**
Age	Minimum − 1	17
Age	Minimum	18
Age	Minimum + 1	19
Age	Maximum − 1	99
Age	Maximum	100
Age	Maximum + 1	101
Weight	Minimum − 1	29 kg
Weight	Minimum	30 kg
Weight	Minimum + 1	31 kg
Weight	Maximum − 1	199 kg
Weight	Maximum	200 kg
Weight	Maximum + 1	201 kg

This ensures that the system handles values **just below, exactly at, and just above the accepted limits**.

8. Decision Table Testing

Scenario: Personalized Meal Recommendation
Conditions:

- C1 = Vegetarian
- C2 = Food allergy
- C3 = Weight-loss goal
- C4 = Valid health information

Rule	**C1 Vegetarian**	**C2 Allergy**	**C3Weight Loss**	**C4 Valid Data**	**Expected Action**
R1	Yes	No	Yes	Yes	Vegetarian weight-loss plan
R2	Yes	Yes	Yes	Yes	Vegetarian plan excluding allergen
R3	No	Yes	Yes	Yes	Normal plan excluding allergen
R4	No	No	Yes	Yes	Weight-loss plan
R5	Yes	No	No	Yes	General vegetarian plan
R6	No	No	No	Yes	General balanced plan
R7	Any	Any	Any	No	Ask user to complete required data

Example:
If the user is vegetarian, has a peanut allergy, and wants weight loss, the system should generate a **vegetarian weight-loss meal plan that does not contain peanuts**.

9. State Transition Testing

The platform can be tested using different user states.

User Account States
Registered → Logged In → Profile Completed → Plan Generated → Meal Tracking → Progress Tracking → Logged Out

Current State	**Event**	**Next State**	**Expected Result**	**Current State**
Unregistered	Register	Registered	Account created	Unregistered
Registered	Login	Logged In	Dashboard displayed	Registered

Logged In	Enter health data	Profile Completed	Profile saved	Logged In
Profile Completed	Generate plan	Plan Generated	Meal plan displayed	Profile Completed
Plan Generated	Record food	Meal Tracking	Food added	Plan Generated
Meal Tracking	View progress	Progress Tracking	Progress displayed	Meal Tracking
Logged In	Logout	Logged Out	Session terminated	Logged In
Logged Out	Access dashboard	Logged Out	Access denied	Logged Out

10. Non-Functional Test Cases

TC ID	Test Area	Test Steps	Expected Result
NFT01	Performance	Generate nutrition plan	Response should be within specified system response time
NFT02	Usability	Navigate through dashboard	User should easily locate major functions
NFT03	Security	Attempt unauthorized profile access	Access should be denied
NFT04	Security	Logout and press Back	Protected information should not be accessible
NFT05	Compatibility	Open system on Chrome, Edge and Firefox	Core functions should work correctly
NFT06	Mobile compatibility	Open platform on smartphone	Layout should adapt correctly
NFT07	Reliability	Perform repeated meal entries	System should not crash or lose data
NFT08	Availability	Access system during normal operating period	System should be accessible
NFT09	Scalability	Simulate multiple simultaneous users	System should remain stable
NFT10	Data accuracy	Compare calculated nutrition totals with expected values	Values should be accurate

11. Traceability Matrix

Requirement	Related Scenarios	Test Cases
FR01 Registration/Login	TS01, TS02	TC01–TC06
FR02 Health Information	TS03, TS04	TC07–TC15
FR03 Dietary Preferences	TS06	TC17–TC18
FR04 Fitness Goals	TS07	TC19
FR05 AI Analysis	TS08	TC20–TC22
FR06 Meal Plans	TS08	TC20–TC22
FR07 Dietary Monitoring	TS09, TS10	TC23–TC27
FR08 Wearable Integration	TS11, TS12	TC28–TC31
FR09 Recommendations	TS13	TC32–TC33
FR10 Progress Tracking	TS14	TC34–TC35
FR11 Profile Updates	TS15	TC36
FR12 Notifications	TS16	TC37–TC38
FR13 Security	TS19	TC39–TC40
FR14 Error Handling	TS17, TS18	TC21, TC31, TC41–TC42
NFR01 Performance	TS20	TC43–TC44
NFR02 Usability	TS20	TC45
NFR03 Reliability	TS20	NFT07
NFR04 Security	TS19	NFT03–NFT04
NFR05 Compatibility	TS20	NFT05–NFT06
NFR06 Scalability	TS20	NFT09
NFR08 Accuracy	TS20	NFT10

12. Test Coverage Summary

Testing Area	Coverage
Functional Requirements	High
Non-Functional Requirements	High
Normal Conditions	✓
Boundary Conditions	✓
Invalid Conditions	✓
Exceptional Conditions	✓
Equivalence Partitioning	✓
Boundary Value Analysis	✓
Decision Table Testing	✓
State Transition Testing	✓
Security Testing	✓
Performance Testing	✓
Usability Testing	✓
Compatibility Testing	✓
Traceability	✓

Coverage Formula
You can calculate requirement coverage using:
Requirement Coverage = (Number of requirements covered by test cases / Total number of requirements) × 100
For example, if all **14 functional requirements** are mapped to at least one test case:
Requirement Coverage = (14 / 14) × 100 = 100%
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13. Overall Test Result
The proposed test suite covers the major functional and non-functional requirements of the **AI-Powered Personalized Nutrition and Health Recommendation Platform**. It includes normal, boundary, invalid, and exceptional conditions and applies **Equivalence Partitioning, Boundary Value Analysis, Decision Table Testing, and State Transition Testing**.
The test cases provide traceability from **requirements → scenarios → test cases**, helping ensure adequate coverage of the system's major functions such as personalized meal generation, health-data management, dietary tracking, wearable integration, recommendations, progress monitoring, security, and performance.